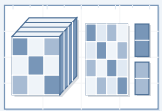
ARC's K=3 / K=4 cumulant recursion (Wu et al. 2026)

a

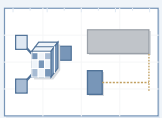
Regime 3: full-scale factored K=4 (§2, §5)



Store (x, y, B, Q), $Q = T_l \cdots T_{b+1}$ **($n \times n$, 1 n^3/layer)**

Static host kernel B_r ($n \times n \times R$), written once

~256 GB → 120 GB host + 60.4 GB GPU peak

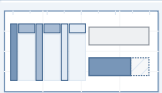


$S_l = \mathrm{Sym}(\sum_r (Qx) \otimes (Qy) \otimes QB_rQ^\top)$, **0 approx.**

Prop. 1; 1024×16 MSE 1.72e-9 vs 2.03e-8

Width law $n^{-3.95}$ vs $n^{-3.08}$

11.8×

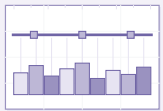


Rank-r matrix pairs: 32 pairs MSE 6.95e-9

525,045 vs dense 756,620 units (−31%)

b

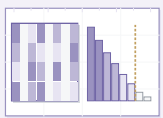
Regime 2: in-budget K=3 chain B (§4)



504aldo V29 (adj 5.418e-9) + Puffi SSC closure

9 closure coefficients refit end to end

Plus Strassen joins, SVD ranks, wall-clock blocks

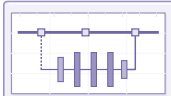


Trim columns $w_1 < 0.014$, young atoms 99.5% s_q

#1 of 8 write-ups' chains w/o corrector/table

adj 3.494e-9

#335301



+ 54,561-param read-only corrector

adj 3.145e-9, #1 of 8, −5.9% vs #335413

#336537